Invariant Relational Selection and Approximate Symmetry

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Abstract

A recurring uncertainty in attempts to infer physical structure from relational data is whether an inferred decomposition is intrinsic or has been inserted through a chosen representation. We sharpen a finite-dimensional negative-control analysis into a classification and a quantitative stability test. For a given finite tuple of self-adjoint operators, an unordered complete rank-one projection family invariant under all data-preserving unitaries exists exactly when the generated unital (*)-algebra acts with multiplicity one in every irreducible block. Such resolutions are unique under the symmetry requirement only in the maximal-abelian case; an irreducible algebra does not suffice for uniqueness. When one operator has simple spectrum, a second operator produces a coordinate-free weighted support relation and gauge-invariant cycle data. We prove that the generated algebra detects connected components but loses all distinction among different connected support graphs, and we establish reconstruction of a connected Hermitian matrix from edge magnitudes and a fundamental set of complex cycle products up to diagonal phase gauge. Finally, a nonnegative commutator Laplacian gives a representation-invariant measure of approximate symmetry. A dense two-generator example has the full matrix algebra for every positive perturbation, while its commutator spectral gap vanishes quadratically in perturbation strength. We derive the leading coefficient and provide explicitly reproducible finite tests. These are mathematical implications of explicitly supplied operator data, not a derivation of such data, a clock, quantum ontology, locality, or spacetime from the prephysical Reality (R) proposal.

Keywords: representation invariance, finite-dimensional commutant, relational reconstruction, approximate symmetry, gauge loop invariants, operational identifiability.

1. Scope and the non-importation requirement

The prephysical Reality (R) proposal currently refers to signed sectors and unspecified relational information, without a demonstrated map to physical Hilbert spaces, CAR generators, a measurement basis, physical clocks, or metric data. No such map is assumed here. All operators below belong to explicitly declared *mathematical test classes*. A theorem about these classes does not, without a separate correspondence theorem, become a theorem about the prephysical proposal or the actual universe.

For a finite ordered tuple $\mathcal{D} = (A_1, \dots, A_k)$ of self-adjoint operators on a finite-dimensional complex Hilbert space $\mathcal{H}$, let

$$\mathcal{A} = \mathrm{Alg}^{*}(I, A_1, \dots, A_k), \qquad G_{\mathcal{D}} = \{U \in U(\mathcal{H}) : UA_jU^{*} = A_j \text{ for all } j\}.$$

A *complete rank-one resolution* is an unordered family $\mathcal{P} = \{P_1, \dots, P_N\}$ of mutually orthogonal rank-one projections with $\sum_i P_i = I$. Being intrinsic to the datum at minimum requires that the family respect changes of unitary representation. This is an inference test, not a postulate about nature.

2. The exact trichotomy of invariant rank-one selection

Theorem 1 (multiplicity trichotomy). There exist integers $n_\alpha, m_\alpha \geq 1$ and a unitary decomposition in which

$$\mathcal{H} \simeq \bigoplus_\alpha (\mathbb{C}^{n_\alpha} \otimes \mathbb{C}^{m_\alpha}), \qquad \mathcal{A} \simeq \bigoplus_\alpha (M_{n_\alpha}(\mathbb{C}) \otimes I_{m_\alpha}),$$

$$\mathcal{A}' \simeq \bigoplus_\alpha (I_{n_\alpha} \otimes M_{m_\alpha}(\mathbb{C})).$$

An unordered complete rank-one resolution invariant as a set under $G_{\mathcal{D}}$ exists if and only if $m_\alpha = 1$ for every α. Subject only to this symmetry-invariance requirement:

1. If some $m_\alpha > 1$, no such resolution exists.
2. If all $m_\alpha = 1$ and at least one $n_\alpha > 1$, infinitely many invariant resolutions exist.
3. If all $n_\alpha = m_\alpha = 1$, exactly one invariant resolution exists, namely the joint minimal spectral projectors of the maximal abelian algebra $\mathcal{A}$.

Proof. The finite-dimensional C^*-algebra decomposition gives the two displayed normal forms. The unitary stabilizer equals the unitary group of the commutant, $G_{\mathcal{D}} = U(\mathcal{A}') \simeq \prod_\alpha U(m_\alpha)$, and is connected. Continuous action on a finite, unordered projector family must fix each projector individually. The fixed-point algebra of $G_{\mathcal{D}}$ equals $(\mathcal{A}')' = \mathcal{A}$. Hence every invariant rank-one projector belongs to $\mathcal{A}$.

Every nonzero projection in block $M_{n_\alpha}(\mathbb{C}) \otimes I_{m_\alpha}$ has rank a positive multiple of m_α. Therefore a complete rank-one resolution is possible precisely when all multiplicities equal one. In that case, all rank-one resolutions within each block are fixed by the block-scalar stabilizer. If any block size $n_\alpha \geq 2$, continuously many basis choices inside that block satisfy the same requirement; if every block size is one, the central minimal projections are the unique complete resolution. □

Qualification. This is a classification of the *stabilizer-invariance requirement*, not of all conceivable inference rules. Additional independently justified structure or criteria may select among the infinitely many invariant resolutions in Case 2. Neither mathematical rank-one selection nor a unique joint eigenbasis implies a physical spatial-mode or many-body tensor-product interpretation.

Corollary 2 (center information). The minimal central projections of $\mathcal{A}$ give a unique unordered family of central blocks, even if Theorem 1 forbids or fails uniquely to select rank-one modes. These blocks are algebraic sectors, not automatically spatial regions.

Proof. The center has the canonical form $\mathcal{Z}(\mathcal{A}) \simeq \bigoplus_\alpha \mathbb{C} I_{n_\alpha m_\alpha}$; its nonzero minimal projections are uniquely determined. □

3. Coordinate-free weighted relations from an independently supplied operator pair

Theorem 3 (spectral support and connected components). Let $D = D^*$ have distinct eigenvalues $d_1, \ldots, d_N$ and corresponding rank-one projectors P_i. Let $B = B^*$ be a second operator. Set

$$w_{ij} = \mathrm{Tr}(P_i B P_j B) = \|P_i B P_j\|_{\mathrm{HS}}^2 \geq 0 \quad (i \neq j).$$

The weights and the resulting undirected support graph are invariant under simultaneous unitary conjugation of (D, B), up to consistent relabeling of its spectral projectors. If its connected components are $C_1, \ldots, C_r$, then

$$\boxed{\mathrm{Alg}^*(D, B) \cong \bigoplus_{\alpha=1}^{r} M_{|C_\alpha|}(\mathbb{C}).}$$

In particular, the joint commutant has complex dimension r and consists of blockwise scalars; the center of the generated algebra likewise has dimension r.

Proof. Spectral interpolation gives

$$P_i = \prod_{j \neq i} \frac{D - d_j I}{d_i - d_j} \in \mathrm{Alg}^*(D, B).$$

For each support edge, $P_i B P_j = B_{ij} |i\rangle\langle j|$ is a nonzero multiple of the matrix unit E_{ij}. Products of adjacent matrix units along any path produce E_{ij} for its endpoints. Hence all matrix units within each connected component lie in the generated algebra. The two original operators have no matrix elements between distinct components, so no cross-component matrix units lie in the algebra. The block structure and commutant conclusion follow. Simultaneous unitary conjugation preserves $\mathrm{Tr}(P_i B P_j B)$. □

Interpretive warning. This does not stipulate a spatial graph as an ontology. It proves that when an independently justified mathematical pair (D, B) is provided, a support relation can be extracted invariantly. It does *not* establish why Reality (R) should provide this pair, why D should distinguish physical modes, or why graph distance should equal physical distance.

Corollary 4 (maximal algebra fails to discriminate connected geometries). Given the same simple-spectrum D, a connected nearest-neighbor chain and a densely coupled complete graph both generate $M_N(\mathbb{C})$. Therefore irreducibility and triviality of the center do not distinguish their interaction patterns.

Proof. Both support graphs are connected; apply Theorem 3. □

Theorem 5 (gauge-relative reconstruction from cycle data). In the setting of Theorem 3, assume the support graph is connected. The diagonal entries $\mathrm{Tr}(P_i B)$, nonzero edge magnitudes $\sqrt{w_{ij}}$, and complex cyclic traces

$$W_{i_1 \ldots i_\ell} = \mathrm{Tr}(P_{i_1} B P_{i_2} B \cdots P_{i_\ell} B)$$

for one oriented fundamental cycle basis of the support graph uniquely determine B up to diagonal unitary gauge $B \mapsto U B U^*$ with $U = \sum_i e^{i\theta_i} P_i$.

Proof. Fix any spanning tree. Starting at a root and choosing vertex phases recursively, make every nonzero tree-edge matrix element real positive. The diagonal entries and all edge magnitudes are already fixed. Each non-tree edge closes one fundamental cycle with the tree; the phase of its complex cyclic product determines its edge phase, since all tree-edge factors are now positive and nonzero. The reconstructed entries of B are therefore unique in this gauge. Undoing the gauge shows uniqueness up to the stated diagonal unitary. □

The cyclic quantities are *algebraic relational invariants*. A measurement scheme capable of recovering them would need separately justified physical preparation, observations, and calibration; none is supplied by this theorem. The same result does not yield metric locality, dimensionality, or causal propagation.

4. Irreducibility is discontinuous; approximate symmetry can be quantified

Define the nonnegative *commutator Laplacian* of a finite self-adjoint operator tuple by

$$\mathcal{L}_{\mathcal{D}}(X) = \sum_{j=1}^{k} [A_j, [A_j, X]],$$

acting on the real Hilbert space $\mathfrak{h}_0$ of traceless Hermitian matrices, equipped with the Hilbert–Schmidt norm. Its Rayleigh quotient is

$$\langle X, \mathcal{L}_{\mathcal{D}}(X)\rangle = \sum_j \|[A_j, X]\|_{\mathrm{HS}}^2.$$

Write $\gamma(\mathcal{D})$ for its minimum eigenvalue on $\mathfrak{h}_0$. The eigenvalues are invariant under simultaneous unitary conjugation of the tuple. The gap is positive exactly when the joint commutant consists of scalars, equivalently when the generated complex $(\,)$*-algebra equals the full matrix algebra. The word* Laplacian* here describes an algebraic positive operator on a matrix space; it is **not** a spatial diffusion operator or a physical generator.

Proposition 6 (exact two-dimensional test). For Pauli matrices Z and X, the eigenvalues of $\mathcal{L}_{(Z,\varepsilon X)}$ on all 2×2 complex matrices are

$$\boxed{0, \quad 4\varepsilon^2, \quad 4, \quad 4(1+\varepsilon^2).}$$

For $0 < \varepsilon \leq 1$, the traceless commutator gap is $4\varepsilon^2$, although the generated algebra equals $M_2(\mathbb{C})$ for every nonzero ε.

Proof. Use the orthogonal operator basis I, X, Y, Z. Double commutation with Z multiplies X, Y by 4 and annihilates I, Z; with X it multiplies Y, Z by 4 and annihilates I, X. Add the contributions. □

Theorem 7 (quantified symmetry restoration). Let $Q = Q^* = Q^{-1}$ on $\mathbb{C}^N$, with both signs present. Suppose $D = D^*$ and $\mathrm{Alg}^*(Q, D) = M_N(\mathbb{C})$. Define

$$\gamma_\varepsilon = \min_{X=X^*, \operatorname{Tr} X=0, \|X\|_{\mathrm{HS}}=1} (\|[Q, X]\|_{\mathrm{HS}}^2 + \varepsilon^2 \|[D, X]\|_{\mathrm{HS}}^2).$$

Let

$$\mathcal{K} = \{Y = Y^*, \operatorname{Tr} Y = 0, [Q, Y] = 0\}, \qquad s^2 = \min_{Y \in \mathcal{K}, \|Y\|_{\mathrm{HS}}=1} \|[D, Y]\|_{\mathrm{HS}}^2.$$

Then $s > 0$ and

$$\boxed{\lim_{\varepsilon \downarrow 0} \frac{\gamma_\varepsilon}{\varepsilon^2} = s^2.}$$

In addition, for $0 < \varepsilon \|D\|_{\mathrm{op}} \leq 1$,

$$\boxed{\frac{s^2}{4}\varepsilon^2 \leq \gamma_\varepsilon \leq s^2 \varepsilon^2.}$$

Proof. Irreducibility ensures that no nonzero traceless Hermitian matrix commutes with both Q and D. Compactness of the unit sphere in $\mathcal{K}$ gives $s > 0$.

Decompose any traceless Hermitian $X = Y + Z$ orthogonally, where $Y \in \mathcal{K}$ is block diagonal with respect to the ± 1 eigenspaces of Q and Z is off-block. Then $\|[Q, X]\|_{\mathrm{HS}}^2 = 4\|Z\|_{\mathrm{HS}}^2$. Put $b = \|D\|_{\mathrm{op}}$ and $c = \|[D, X]\|_{\mathrm{HS}}$. By the triangle inequality,

$$s\|Y\|_{\mathrm{HS}} \leq c + 2b\|Z\|_{\mathrm{HS}},$$

so $s^2\|Y\|^2 \leq 2c^2 + 8b^2\|Z\|^2$. With $E = 4\|Z\|^2 + \varepsilon^2 c^2$ and $\varepsilon b \leq 1$, it follows that $E \geq \varepsilon^2 s^2 \|Y\|^2/2$, while also $E \geq 4\|Z\|^2$. Averaging yields $E \geq \varepsilon^2 s^2 \|Y\|^2/4 + 2\|Z\|^2 \geq \varepsilon^2 s^2 \|X\|^2/4$, because $s \leq 2b$. Choosing a unit minimizer Y in $\mathcal{K}$ gives the upper bound $\gamma_\varepsilon \leq \varepsilon^2 s^2$.

For the limit, let $X_\varepsilon = Y_\varepsilon + Z_\varepsilon$ minimize the quotient. The upper bound forces $\|Z_\varepsilon\| \leq \varepsilon s/2$. By finite-dimensional compactness, every null sequence of ε has a subsequence with $X_\varepsilon \to Y \in \mathcal{K}$ and $\|Y\| = 1$. Consequently

$$\liminf \frac{\gamma_\varepsilon}{\varepsilon^2} \geq \liminf \|[D, X_\varepsilon]\|^2 = \|[D, Y]\|^2 \geq s^2.$$

Together with the upper bound this proves the limit. □

Thus full algebraic distinguishability can appear for arbitrarily small ε, while the quantitative gap identifying that new information tends to zero as ε^2. This is a rigorous warning against equating exact commutant rank with stable recovery from uncertain data. No noise distribution or experimental sampling model was assumed.

5. Dense negative control and numerical replication

Let Q_N be a normalized real symmetric Paley conference involution of order $N = p + 1$ for a prime $p \equiv 1 \pmod 4$. In its *declared mathematical representation*, $Q_{ij} \neq 0$ for all $i \neq j$. Set

$$D_N = \operatorname{diag}\left(-1, -1 + \frac{2}{N-1}, \ldots, 1\right).$$

The eigenvalues of D_N are distinct, and Theorem 3 implies

$$\operatorname{Alg}^*(Q_N, \varepsilon D_N) = M_N(\mathbb{C}) \quad (\varepsilon \neq 0),$$

while $\operatorname{Alg}^*(Q_N) = \mathbb{C}I \oplus \mathbb{C}Q_N$ and its commutant has dimension $N^2/2$.

Since $\operatorname{Tr} Q_N = 0$, using $Q_N/\sqrt{N}$ as a test direction gives the explicit analytic bound

$$0 < \gamma_\varepsilon \leq \frac{\varepsilon^2}{N}\|[D_N, Q_N]\|_{\mathrm{HS}}^2 = \boxed{\frac{2N(N+1)}{3(N-1)^2}\varepsilon^2}.$$

The equality follows directly from $|Q_{ij}|^2 = 1/(N-1)$ for $i \neq j$ and the closed formula for the variance of an equally spaced sample on $[-1, 1]$.

The numerical checks use standard double-precision Hermitian eigensolvers. The reduced coefficient s^2 was computed explicitly from the commutant subspace of Q_N; the table checks convergence predicted by Theorem 7.

N	Unperturbed commutant dimension	Dimension for $\varepsilon > 0$	Reduced s^2	$\gamma_{0.1}/0.1^2$	$\gamma_{0.02}/0.02^2$
6	18	1	0.11590365	0.11588314	0.11590283
14	98	1	0.03151170	0.03149121	0.03151088

In a separate $N = 6$ check, a chain, a complete Paley support, and three disconnected pairs were examined with the same simple-spectrum diagonal D. Their support graphs have 5, 15, and 3 edges respectively. The algebraic commutant dimensions were 1, 1, and 3, agreeing with the component theorem. An explicit complex diagonal rephasing preserved triangular traces within floating-point error; changing one edge phase with its magnitude fixed changed a triangle trace, as required.

Numerical status. All formula-derived inequalities and the 2×2 eigenvalues were explicitly checked. Floating-point residuals and convergence do not constitute external validation, exact algebraic certificates, or a proof of novelty. The theorems above are proved analytically; computation is an additional consistency check.